Research Statement

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My research lies at the intersection of singularity theory and low-dimensional topology. It is known that Milnor open books of isolated complex singularities always support tight contact structures. During my Ph.D. studies I have focused on the question of whether overtwisted contact structures also arise from singularity theory, using real analytic maps. Here, I summarize the results obtained jointly with my advisor, F. Öztürk, the techniques we used and future directions around the topic.

Remark. AI Declaration. The content of the research and proofs mentioned here is obtained by the author and her Ph.D. advisor and does not include ideas and discussions by AI. The author uses Claude AI as an assistant; for discussions to verify her knowledge, double checking technical computations and sometimes for proof-reading or grammar check.

1 Summary of Research

For a holomorphic function $f : (\mathbb{C}^2, 0) \to (\mathbb{C}, 0)$ with an isolated singularity at the origin, one has the smooth link $L_f = f^{-1}(0) \cap S^3_\varepsilon$ together with the fibration $f/|f| \colon S^3_\varepsilon \setminus L_f \to S^1$ so that the pair $(L_f, f/|f|)$ is an open book decomposition of S^3 by Milnor's fibration theorem [1]. The complement of the link L_f is fibered by surfaces, so called the pages,each bounded by L_f, so called the binding. As an interaction with contact topology, W. P. Thurston and H. E. Winkelnkemper showed that every open book supports a contact structure [2], and E. Giroux showed that this correspondence is a bijection between open books up to positive stabilization and contact structures up to isotopy [3]. A contact structure is a plane field that is nowhere tangent to any surface. They come in two types; a contact structure is *overtwisted* if it contains an embedded disc whose boundary is tangent to the contact planes, and *tight* otherwise. In this way, every Milnor open book determines a contact structure, and this is always the standard tight one on S^3.

This theory for open book decompositions arising from Milnor's fibration theorem on complex plane was generalized by C. Caubel, A. Némethi, P. Popescu-Pampu to normal surface singularities [4]. For a normal complex surface singularity $(X, 0)$ embedded in some $\mathbb{C}^N$, the intersection $M_{X,\varepsilon} = X \cap S^{2N-1}_\varepsilon$, called the link manifold (independent of small $\varepsilon > 0$, so we drop ε in the notation), is a smooth 3-manifold endowed with a canonical contact structure $\xi_{\text{can}} = TM_X \cap J(TM_X)$ given by the complex tangencies. Given a holomorphic function $f : (X, 0) \to (\mathbb{C}, 0)$ with an isolated singularity at the origin, $L_f = f^{-1}(0) \cap M_X$ with $\pi = f/|f| : M_X \setminus L_f \to S^1$ defines an open book decomposition of M_X, called a *Milnor open book*. For $(X, 0) = (\mathbb{C}^2, 0)$, this definition recovers Milnor's fibration theorem. Moreover, in [4], it is shown that the contact structure supported by a Milnor open book is the canonical contact structure of the ambient manifold and this structure is Stein fillable, hence tight.

The main question we have focused is whether we can find algebraic realization of overtwisted contact structures in similar sense. By Y. Eliashberg, overtwisted contact structures on a closed

3-manifold M are classified, up to isotopy, by the homotopy class of plane fields [5]. When M is a rational homology sphere, this homotopy class is determined by the induced spin^c structure and the 3-dimensional invariant $d_3(\xi)$ [6]. As a first step, we restricted to S^3 which has a unique spin^c structure. There are countably infinitely many overtwisted contact structures, classified up to isotopy by their 3-dimensional invariant, $d_3 \in \mathbb{Z} + \frac{1}{2}$ and the standard contact structure is the unique tight one.

The natural direction to consider is given in the last chapter of Milnor's book [1, Chapter 11], where the Milnor's fibration theorem is extended to real analytic maps $h : (\mathbb{R}^4, 0) \to (\mathbb{R}^2, 0)$ which have an isolated critical point at the origin. In this case, $L_h = h^{-1}(0) \cap S^3_\varepsilon$ is a link in S^3_ε and links arising in this way are called *real algebraic links*. A *real algebraic open book decomposition* on the 3-sphere is by definition the Milnor fibration of such a real algebraic function. Note that in the real setting, the critical point set can be 1-dimensional unlike the complex case, therefore the isolated critical point hypothesis is too restrictive to find examples. When this condition holds, Milnor's fibration theorem gives a fibration but it is not necessarily given by the argument map $\frac{h}{|h|}$. So the fibration might not determine an honest open book decomposition. K. Inaba showed that every contact structure on S^3 is supported by a real algebraic open book. However, in his construction, the pages of open books have various genera. It is known that every overtwisted contact structure is supported by a planar open book [7], so we refine his result as follows.

Theorem 1 (K., Öztürk, 2025)**.** *Every overtwisted contact structure on S^3 is supported by a real algebraic open book with planar pages.*

The proof of the theorem is constructive. We explicitly exhibit real algebraic polynomials whose Milnor fibrations realize every value of d_3.

- In earlier work [8], we start with infinite families of Seifert links in S^3 given in [9]. Using the underlying Seifert structure, we compute the monodromy of the open books and show that open books have planar pages. Then we conclude that the supported contact structures are overtwisted by computing the d_3 invariant.
- For each of these links, we construct a map of the form $f\bar{g}$ with an isolated singularity at the origin, which is real algebraic by [10]. We see that the link of the map is the given Seifert link by resolution of the singularity.
- To obtain a wider class of links, we do suitable splicing operations that preserve algebraicity and together with their spliced links, these family of links realize all but finitely many positive values of d_3.
- To compute the d_3 invariant of the supported contact structure, we use the page and monodromy of the open book to construct an almost complex 4-manifold bounded by S^3 with given open book, and compute d_3 from its intersection form.
- The remaining values of d_3 require considering slightly different families of real algebraic maps. A fourth family of Seifert links given in [9] realizes all positive values of d_3, but since the action of the underlying circle has fixed points, these links cannot be realized by maps of the form $f\bar{g}$. We construct a polar weighted homogeneous polynomial realizing this family of links and show that it is real algebraic.
- We realize that these links can be obtained by a sequence of negative and positive Hopf plumbing on open book of the unknot therefore, the number of negative link components determines the value of d_3 invariant.

- For negative values, mirroring the Seifert link on S^3 corresponds to conjugating one variable of the real analytic function defining it, so the first three families remain real algebraic, and we splice their mirrors with the fourth family of links in a way to preserve algebraicity. Mirroring gives negative values of d_3 invariant and as splicing with the fourth family again corresponds to Hopf plumbing on the spliced open book in our case, so we can realize all negative values.

We then extend this result to Milnor fillable 3-manifolds as in the complex set up. A contact 3-manifold is called *Milnor fillable* if it is contactomorphic to the link $M_X = X \cap S_\varepsilon^{2N-1}$ of an isolated complex surface singularity $(X, 0) \subset (\mathbb{C}^N, 0)$ equipped with its canonical contact structure. Instead of holomorphic functions on $(X, 0)$, we consider complex-valued real analytic functions $h : (X, 0) \to (\mathbb{C}, 0)$ and give the following definition.

Definition 1. Let $(X, 0)$ be a normal complex surface singularity with link manifold M_X, and let $h : (X, 0) \to (\mathbb{C}, 0)$ be a complex-valued real analytic germ. Consider $L_h = h^{-1}(0) \cap M_X$ and

$$\phi_h = h/|h| : M_X \setminus L_h \to S^1.$$

We call the pair (L_h, ϕ_h) a *weakly real Milnor open book* associated with h provided that L_h is a smooth link, ϕ_h is a smooth locally trivial fibration, and for each component K_i of L_h there exists a trivialization $S^1 \times D^2 \cong \nu(K_i)$ and integers $n_i \in \mathbb{Z} \setminus \{0\}$, $k_i \in \mathbb{Z}$ with $\phi_h|_{\nu(K_i)\setminus K_i} = (z/|z|)^{n_i} t^{k_i}$ where $t \in S^1$ and $z \in D^2$ denote the coordinates of the two factors. If moreover h has no critical point on $X \setminus \{0\}$ near 0, we call the pair a *real Milnor open book*. We say that (L_h, ϕ_h) is integral if $|n_i| = 1$ for all i (corresponding to an honest open book), and otherwise rational in the sense of K. Baker, J. Etnyre, J. Van Horn-Morris [11].

This definition covers the definition and examples of open books obtained by Milnor's fibration theorem and generalizes the notion of Milnor open books. Examples of weakly real Milnor open books appear in the literature given by A. Pichon and J. Seade [12].

This leads to the following question.

Question 1. Is it true that every overtwisted contact structure on a Milnor fillable 3-manifold is supported by a (weakly) real Milnor open book?

We call a contact structure real algebraic if it is supported by a (weakly) real Milnor open book. We extend the constructions we have done for S^3 to lens spaces, which arise both as quotients of the 3-sphere under finite cyclic group actions and as links of cyclic quotient surface singularities. We prove the following.

Theorem 2 (K., Öztürk, 2026)**.** *Every lens space $L(p, q)$ carries infinitely many nonisotopic real algebraic overtwisted contact structures, supported by planar real Milnor open books which are rational. Moreover, on $L(p, p-1)$ these can be taken to be honest open books, and in this case, all possible values for d_3 are covered.*

The constructions in the proof passes to lens spaces through the covering $S^3 \to L(p, q) = S^3/\mathbb{Z}_p$.

- The idea builds on E. Horvat's notion of algebraic links in lens spaces [13], which we first corrected accordingly for lifting criteria given by E. Manfredi in [14]. Then we extend the definition to mixed functions.

- A mixed polynomial is (p,q)-*invariant* if the $\mathbb{Z}_p$-action multiplies it by ζ^k. Its zero set is invariant under the action and some k power of its argument map descends to a fibration on $L(p,q)$. The induced open book is honest when $k=0$ and rational otherwise. We see that planarity is preserved by computing Euler characteristic for the lifting.
- The fourth family of polynomials we have constructed in S^3 case are (p,q)-invariant precisely when $k=1+q$. Therefore, we obtain open honest books on $L(p,p-1)$.
- For the mirrored families, we choose coefficients so that the action permutes the factors cyclically, which makes their splices with the last family invariant as well.
- To show that the induced contact structures are overtwisted, we use M. Ishikawa's compatible contact form for Seifert multilinks [15] being $\mathbb{Z}_p$-invariant, so it descends to $L(p,q)$ together with its Lutz tube.
- To distinguish them, we combine M. Bou Khuzam's lifting formula [16] with the G-signature theorem, computing the signature defect of the covering through Hirzebruch's Dedekind sum formula. This yields an explicit formula for the d_3 invariant of the descended contact structures in terms of d_3 invariant of their lifting and Dedekind sum.

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